

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	J. MEHER KOUSHIK	Reg. No.:	192525211
Section / Batch:	AI&ML	Date:	24/07/2026

Code of Conduct

I J. MEHER KOUSHIK (192525211) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: J. MEHER KOUSHIK

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25

Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

1.



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Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

```

}
return -1;
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 int binarySearch(int arr[], int n,int key) {
2     int low = 0, high = n - 1;
3     while(low <= high){
4         int mid = low + (high - low) / 2;
5         if(arr[mid] == key)
6             return mid;
7         else if(arr[mid] < key)
8             low = mid + 1;
9         else
10            high = mid - 1;
11    }
12    return - 1;
13 }
14 int main() {
15     int arr[] = {10,20,30,40,50};
16     int n = 5;
17     int key = 40;
18     int result = binarySearch(arr,n,key);
19     if(result != -1){
20         printf("element found %d\n",result);
21     } else {
22         printf("not found");
23     }
24 }

```

Test Input
10 20 30 40 50
5
30

Output
element found 3

2



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Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Find the error in the stack deletion operation. Justify it.

```

int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 int stack[5] = {10, 20, 30};
3 int top = 2;
4 int pop() {
5     if (top == -1) {
6         printf("Underflow\n");
7         return -1;
8     }
9     return stack[top--];
10 }
11 int main() {
12     int x;
13     x = pop();
14     if (x != -1)
15         printf("Popped element: %d", x);
16     return 0;
17 }

```

Test Input
10 20 30
2

Output
Popped element: 30

3.



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QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List 5 marks

Q3 Stack 5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int stack[5];
3 int top;
4 void init() {
5     top = -1;
6 }
7 int main() {
8     init();
9     printf("Top = %d\n", top);
10    return 0;
11 }
```

Test Input

stdin input (optional)

Output

Top = -1

Run

Save

Submit Fix

4.



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QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List 5 marks

Q9 Linked List 5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
    while(stack[top] != ')') {
        postfix[j++] = pop();
    }
    pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 char stack[10] = {'('};
3 int top = 0;
4 char postfix[20];
5 int j = 0;
6 char pop() {
7     return stack[top--];
8 }
9 int main() {
10    char ch = ')';
11    if (ch == ')') {
12        while (stack[top] != '(') {
13            postfix[j++] = pop();
14        }
15        pop();
16    }
17    printf("Program executed successfully");
18    return 0;
19 }
```

Test Input

(A+B)*C

Output

Program executed successfully

Run

Save

Submit Fix

5.



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QUESTIONS

Q1
 Binary Search
 5 marks

Q2
 Stack
 5 marks

Q3
 Stack
 5 marks

Q4
 Stack
 5 marks

Q5
 Stack - Balancing Symbols
 5 marks

Q6
 Singly Linked List
 5 marks

Q7
 Stack
 5 marks

Q8
 Linked List
 5 marks

Q9

Q5 Stack - Balancing Symbols 5 marks

In balancing Bracket, you identify the error.

```

if((ch == '[' && stack[top] == '[') ||
(ch == ']' && stack[top] == '[')) {
    pop();
} else {
    return 0;
}
    
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 char stack[10] = {'['};
3 int top = 0;
4 void pop() {
5     top--;
6 }
7 int main() {
8     char ch = '[';
9     if ((ch == '[' && stack[top] == '[') ||
10        (ch == ']' && stack[top] == '[')) {
11         pop();
12         printf("Balanced Brackets");
13     } else {
14         printf("Not Balanced");
15     }
16     return 0;
    
```

Test Input

```

{[{}]}
    
```

Output

```

Balanced Brackets
    
```

6.



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QUESTIONS

Q2
 Stack
 5 marks

Q3
 Stack
 5 marks

Q4
 Stack
 5 marks

Q5
 Stack - Balancing Symbols
 5 marks

Q6
 Singly Linked List
 5 marks

Q7
 Stack
 5 marks

Q8
 Linked List
 5 marks

Q9
 Linked List
 5 marks

Q10
 Queue
 5 marks

```

while temp.next != NULL:
    print(temp.data)
    temp = temp.next
    
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #include <stdlib.h>
3 struct Node {
4     int data;
5     struct Node *next;
6 };
7 int main() {
8     struct Node *head, *second, *third, *temp;
9     head = (struct Node*)malloc(sizeof(struct Node));
10    second = (struct Node*)malloc(sizeof(struct Node));
11    third = (struct Node*)malloc(sizeof(struct Node));
12    head->data = 10;
13    head->next = second;
14    second->data = 20;
15    second->next = third;
16    third->data = 30;
17    third->next = NULL;
18    temp = head;
19    while (temp != NULL) {
20        printf("%d ", temp->data);
21        temp = temp->next;
22    }
23    return 0;
24 }
    
```

Test Input

```

stdin input (optional)
    
```

Output

```

10 20 30
    
```

7.



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Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2
3 int stack[5] = {10, 20, 30};
4 int top = 2;
5
6 int pop() {
7     if (top == -1) {
8         printf("Underflow\n");
9         return -1;
10    }
11    return stack[top--];
12 }
13
14 int main() {
15     int x;
16
17     x = pop();
18
19     if (x != -1)
20         printf("Popped element = %d", x);
21
22     return 0;
23 }

```

Test Input
{10, 20, 30}
2

Output
Popped element = 30

8.



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Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

```

1 #include <stdio.h>
2 #include <stdlib.h>
3 struct Node {
4     int data;
5     struct Node* next;
6 };
7 int main() {
8     struct Node n1 = {10, NULL};
9     struct Node n2 = {20, NULL};
10    struct Node n3 = {30, NULL};
11    n1.next = &n2;
12    n2.next = &n3;
13    struct Node* current = &n1;
14    struct Node* prev = NULL;
15    struct Node* next = NULL;
16    while (current != NULL) {
17        next = current->next;
18        current->next = prev;
19        prev = current;
20        current = next;
21    }
22    struct Node* temp = prev;
23    while (temp != NULL) {
24        printf("%d -> ", temp->data);
25        temp = temp->next;
26    }
27    printf("NULL\n");
28    return 0;
29 }
30

```

Test Input
10 -> 20 -> 30 -> NULL

Output
30 -> 20 -> 10 -> NULL

10/10 answered

9.



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Linked List
5 marks

Q1 Binary Search 5 marks ✓

Q2 Stack 5 marks ✓

Q3 Stack 5 marks ✓

Q4 Stack 5 marks ✓

Q5 Stack - Balancing Symbols 5 marks ✓

Q6 Singly Linked List 5 marks ✓

Q7 Stack 5 marks ✓

Q8 Linked List 5 marks ✓

Q9 Linked List 5 marks ✓

What is the error in the below code?

```
new.next = head
head = new
```

✳ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head = NULL;
11     struct Node *newNode;
12
13     newNode = (struct Node *)malloc(sizeof(struct Node));
14     newNode->data = 10;
15
16     newNode->next = head;
17     head = newNode;
18
19     printf("Inserted element: %d\n", head->data);
20
21     return 0;
22 }
```

Test Input

10

Output

Inserted element: 10

10.



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Binary Search 5 marks ✓

Q2 Stack 5 marks ✓

Q3 Stack 5 marks ✓

Q4 Stack 5 marks ✓

Q5 Stack - Balancing Symbols 5 marks ✓

Q6 Singly Linked List 5 marks ✓

Q7 Stack 5 marks ✓

Q8 Linked List 5 marks ✓

Q9 Linked List 5 marks ✓

Q10 Linked List 5 marks ✓

What is missing in the below pseudocode?

```
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]
```

✳ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int Q[5] = {10, 20, 30};
3 int front = 0, rear = 2;
4 int dequeue() {
5     if (front == -1) {
6         printf("Underflow\n");
7         return -1;
8     }
9     int item = Q[front++];
10    if (front > rear) {
11        front = rear = -1;
12    }
13    return item;
14 }
15 int main() {
16     int x;
17     x = dequeue();
18     if (x != -1)
19         printf("Deleted element = %d", x);
20     return 0;
21 }
```

Test Input

{10, 20, 30}
0
2

Output

Deleted element = 10

11.

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QUESTIONS

- Q1** Stack 5 marks ✓
- Q2** Circular Queue 5 marks ✓
- Q3** Stack 5 marks ✓
- Q4** Stack 5 marks ✓
- Q5** Doubly Linked List 5 marks ✓
- Q6** Queue 5 marks ✓
- Q7** Queue - Deque 5 marks ✓
- Q8** Binary Search 5 marks ✓
- Q9** ✓

Q1 Stack 5 marks

Point out the error?

```
PEEK(stack):
return stack[top+1]
```

✳ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int stack[5] = {10, 20, 30};
3 int top = 2;
4 int peek() {
5     if (top == -1) {
6         printf("Stack is Empty\n");
7         return -1;
8     }
9     return stack[top];
10 }
11 int main() {
12     int x;
13     x = peek();
14     if (x != -1)
15         printf("Top element = %d", x);
16     return 0;
17 }
```

Test Input

```
{10, 20, 30}
2
```

Output

```
Top element = 30
```

Run Save Submit Fix

12.

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
[← Back](#)
QUESTIONS

- Q1** Stack 5 marks ✓
- Q2** Circular Queue 5 marks ✓
- Q3** Stack 5 marks ✓
- Q4** Stack 5 marks ✓
- Q5** Doubly Linked List 5 marks ✓
- Q6** Queue 5 marks ✓
- Q7** Queue - Deque 5 marks ✓
- Q8** Binary Search 5 marks ✓

Q2 Circular Queue 5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}
```

✳ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 5
3 int queue[MAX];
4 int front = 0, rear = 2;
5 int vehicle = 40;
6 int main() {
7     if ((rear + 1) % MAX == front) {
8         printf("Queue Full");
9     } else {
10         rear = (rear + 1) % MAX;
11         queue[rear] = vehicle;
12         printf("Vehicle inserted = %d", vehicle);
13     }
14     return 0;
15 }
```

Test Input

```
5
0
2
40
```

Output

```
Vehicle inserted = 40
```

13.



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CO2_A12_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8

Q3 Stack 5 marks

Point out the error.
 if top == MAX:
 Overflow

✨ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #define MAX 5
3 int stack[MAX];
4 int top = 4;
5 int item = 50;
6 int main() {
7     if (top == MAX - 1) {
8         printf("Overflow");
9     } else {
10        stack[++top] = item;
11        printf("Inserted element = %d", item);
12    }
13    return 0;
14 }

```

Run Save Submit Fix

Test Input

5
4
50

Output

Overflow

14.



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QUESTIONS

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q4 Stack 5 marks

Identify the issue in the below pseudocode.
 POP():
 if top == 0 → Underflow

✨ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 int stack[5] = {10, 20, 30};
3 int top = 2;
4 int pop() {
5     if (top == -1) {
6         printf("Underflow\n");
7         return -1;
8     }
9     return stack[top--];
10 }
11 int main() {
12     int x;
13     x = pop();
14     if (x != -1)
15         printf("Popped element = %d", x);
16     return 0;
17 }

```

Run Save Submit Fix

Test Input

{10, 20, 30}
2

Output

Popped element = 30

15.



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0 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

10/10 answered

Your Fixed Code

```

1 #include <stdio.h>
2 #include <stdlib.h>
3 struct Node {
4     int data;
5     struct Node *prev, *next;
6 };
7 void deleteNode(struct Node **head, struct Node *del) {
8     if (del == NULL)
9         return;
10    if (*head == del)
11        *head = del->next;
12    if (del->prev != NULL)
13        del->prev->next = del->next;
14    if (del->next != NULL)
15        del->next->prev = del->prev;
16    free(del);
17 }
18 int main() {
19     struct Node *head, *second;
20     head = (struct Node *)malloc(sizeof(struct Node));
21     second = (struct Node *)malloc(sizeof(struct Node));
22     head->data = 10;
23     head->prev = NULL;
24     head->next = second;
25     second->data = 20;
26     second->prev = head;
27     second->next = NULL;
28     deleteNode(&head, second);
29     printf("Remaining Node: %d", head->data);
30     free(head);
31     return 0;
32 }

```

Test Input

10 <-> 20
20

Output

Remaining Node: 10

16.



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QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.

```

void enqueue(int data) {
    if (rear == MAX) return;
    queue[rear++] = data;
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #define MAX 5
3 int queue[MAX];
4 int rear = 2;
5 void enqueue(int data) {
6     if (rear == MAX - 1) {
7         printf("Queue Overflow\n");
8         return;
9     }
10    queue[++rear] = data;
11 }
12 int main() {
13     enqueue(40);
14     printf("Inserted element = %d", queue[rear]);
15     return 0;
16 }

```

Test Input

2
40

Output

Inserted element = 40

17.



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QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q7 Queue - Deque

5 marks

Find the error in this Deque insertFront operation.

```

void insertFront(int data) {
    if (front == 0) front = MAX - 1;
    else front = front - 1;
    deque[front] = data;
}
      
```

✨ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code


```

1 #include <stdio.h>
2 #define MAX 5
3 int deque[MAX];
4 int front = 2;
5 void insertFront(int data) {
6     if (front == 0)
7         front = MAX - 1;
8     else
9         front = front - 1;
10    deque[front] = data;
11 }
12 int main() {
13    insertFront(10);
14    printf("Inserted element = %d", deque[front]);
15    return 0;
16 }
      
```

Test Input

2
10

Output

Inserted element = 10

18.



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5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

if (arr[mid] > x) return binarySearch(arr, l, mid, x);
 return binarySearch(arr, mid, r, x);
 }

✨ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code


```

1 #include <stdio.h>
2 int binarySearch(int arr[], int l, int r, int x) {
3     if (l > r)
4         return -1;
5     int mid = l + (r - l) / 2;
6     if (arr[mid] == x)
7         return mid;
8     if (arr[mid] > x)
9         return binarySearch(arr, l, mid - 1, x);
10    return binarySearch(arr, mid + 1, r, x);
11 }
12 int main() {
13    int arr[] = {10, 20, 30, 40, 50};
14    int n = 5, x = 40;
15    int result = binarySearch(arr, 0, n - 1, x);
16    if (result != -1)
17        printf("Element found at index %d", result);
18    else
19        printf("Element not found");
20    return 0;
21 }
      
```

Test Input

10 20 30 40 50
40

Output

Element found at index 3

Run

Save

Submit Fix

19.



SIMATS Online Programming Lab
DS PROGRAMMING


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5 marks
Q4
Stack
5 marks
✓

5 marks
Q5
Doubly Linked List
5 marks
✓

5 marks
Q6
Queue
5 marks
✓

5 marks
Q7
Queue - Deque
5 marks
✓

5 marks
Q8
Binary Search
5 marks
✓

5 marks
Q9
Circular Queue
5 marks
✓

5 marks
Q10
Stack - Postfix Expression
5 marks
✓

10/10 answered

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #define MAX 5
3 int queue[MAX] = {10, 20, 30};
4 int front = 0, rear = 2;
5 int dequeue() {
6     if (front == -1) {
7         return -1;
8     }
9     int data = queue[front];
10    if (front == rear) {
11        front = rear = -1;
12    } else {
13        front = (front + 1) % MAX;
14    }
15    return data;
16 }
17 int main() {
18     int x = dequeue();
19     if (x == -1)
20         printf("Queue Underflow");
21     else
22         printf("Deleted element = %d", x);
23     return 0;
24 }

```

Run
Save
Submit Fix

Test Input

```

{10, 20, 30}
0
2

```

Output

```

Deleted element = 10

```

20.



SIMATS Online Programming Lab
DS PROGRAMMING


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5 marks
Q4
Stack
5 marks
✓

5 marks
Q5
Doubly Linked List
5 marks
✓

5 marks
Q6
Queue
5 marks
✓

5 marks
Q7
Queue - Deque
5 marks
✓

5 marks
Q8
Binary Search
5 marks
✓

5 marks
Q9
Circular Queue
5 marks
✓

5 marks
Q10
Stack - Postfix Expression
5 marks
✓

10/10 answered

```

// perform operation: result = op1 + op2;
push(result);
}
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 int stack[10];
3 int top = -1;
4 void push(int x) {
5     stack[++top] = x;
6 }
7 int pop() {
8     return stack[top--];
9 }
10 int main() {
11     push(2);
12     push(3);
13     int op2 = pop();
14     int op1 = pop();
15     int result = op1 + op2;
16     push(result);
17     printf("Result = %d", pop());
18     return 0;
19 }

```

Run
Save
Submit Fix

Test Input

```

23+

```

Output

```

Result = 5

```